

FIGURE 5.1.12 Graph of solution given in (28)

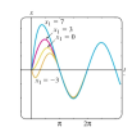


FIGURE 5.1.13 Graph of solution in Example 7 for various x_1

TRANSIENT AND STEADY-STATE TERMS When F is a periodic function, such as $F(t) = F_0 \sin \gamma t$ or $F(t) = F_0 \cos \gamma t$, the general solution of (25) for $\lambda > 0$ is the sum of a nonperiodic function $x_d(t)$ and a periodic function $x_p(t)$. Moreover, $x_d(t)$ dies off as time increases—that is, $\lim_{t \rightarrow \infty} x_d(t) = 0$. Thus for large values of time, the displacements of the mass are closely approximated by the particular solution $x_p(t)$. The complementary function $x_d(t)$ is said to be a **transient term** or **transient solution**, and the function $x_p(t)$, the part of the solution that remains after an interval of time, is called a **steady-state term** or **steady-state solution**. Note therefore that the effect of the initial conditions on a spring/mass system driven by F is transient. In the particular solution (28), $e^{-\gamma t}[\cos t - \frac{1}{2} \sin t]$ is a transient term, and $x_p(t) = -\frac{1}{25} \cos 4t + \frac{1}{25} \sin 4t$ is a steady-state term. The graphs of these two terms and the solution (28) are given in Figures 5.1.12(a) and 5.1.12(b), respectively.

EXAMPLE 7 Transient/Steady-State Solutions

The solution of the initial-value problem

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 2x = 4 \cos t + 2 \sin t, \quad x(0) = 0, \quad x'(0) = 0,$$

where x_1 is constant, is given by

$$x(t) = (x_1 - 2)e^{-t} \sin t + 2 \sin t.$$

Solution curves for selected values of the initial velocity x_1 are shown in Figure 5.1.13. The graphs show that the influence of the transient term is negligible for about $t > 3\pi/2$.

DE OF DRIVEN MOTION WITHOUT DAMPING With a periodic impressed force and no damping force, there is no transient term in the solution of a problem. Also, we shall see that a periodic impressed force with a frequency near or the same as the frequency of free undamped vibrations can cause a severe problem in any oscillatory mechanical system.

Rules to find a complementary function

$$(a_n D^n + a_{n-1} D^{n-1} + a_{n-2} D^{n-2} + \dots + a_0) y = 0 \longrightarrow \textcircled{1}$$

Case 1 $\rightarrow$ If roots are real & distinct ($m_1 \neq m_2$)

$$CF = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \dots + c_n e^{m_n x}$$

Case 2 $\rightarrow$ When roots are real & equal ($m_1 = m_2 = m$)

$$CF = (c_1 + c_2 x) e^{m x} + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

Case 3 $\rightarrow$ When roots are imaginary

$$m_1 = \alpha + i\beta, \quad m_2 = \alpha - i\beta$$

$$CF = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

31. A mass of 1 slug, when attached to a spring, stretches it 2 feet and then comes to rest in the equilibrium position. Starting at $t = 0$, an external force equal to $f(t) = 8 \sin 4t$ is applied to the system. Find the equation of motion if the surrounding medium offers a damping force that is numerically equal to 8 times the instantaneous velocity.
32. In Problem 31 determine the equation of motion if the external force is $f(t) = e^{-t} \sin 4t$. Analyze the displacements for $t \rightarrow \infty$.
33. When a mass of 2 kilograms is attached to a spring whose constant is 32 N/m, it comes to rest in the equilibrium position. Starting at $t = 0$, a force equal to $f(t) = 68e^{-2t} \cos 4t$ is applied to the system. Find the equation of motion in the absence of damping.

(31) $m = 1 \text{ slug}, \quad \gamma = 2 \text{ ft}, \quad \beta = 8$

Here, $F = kx$

$$\Rightarrow 1 \times 32 = k \times 2$$

$$\Rightarrow k = 16$$

$$\left| \begin{array}{l} \frac{\beta}{m} = 2\lambda \\ \Rightarrow \lambda = 4 \end{array} \right.$$

$$\frac{d^2x}{dt^2} + 8 \frac{dx}{dt} + 16x = 8 \sin 4t$$

$$\hookrightarrow m^2 + 8m + 16 = 0$$

$$\Rightarrow (m+4)^2 = 0$$

$$\Rightarrow m = -4, -4$$

$\hookrightarrow$ roots are real and equal.

$$\therefore CF: x_c(t) = e^{-\lambda t} (c_1 + c_2 t)$$

$$\Rightarrow x_c(t) = e^{-4t} (c_1 + c_2 t)$$

$$\therefore PI: x_p(t) = A \sin 4t + B \cos 4t$$

$$x_p'(t) = 4A \cos 4t + 4B \sin 4t$$

$$x_p''(t) = -16A \sin 4t - 16B \cos 4t.$$

$$x'' + 8x' + 16 = f(t)$$

$$\Rightarrow -16A \sin 4t - 16B \cos 4t + 32A \cos 4t + 32B \sin 4t$$

$$16A \sin 4t + 16B \cos 4t = 8 \sin 4t + 0 \cos 4t$$

$$\left. \begin{array}{l} -16A - 32B + 16A = 8 \\ -16B + 32A + 16B = 0 \end{array} \right\} \begin{array}{l} A = 0 \\ B = -\frac{1}{4} \end{array}$$

$$\therefore x_p(t) = -\frac{1}{4} \cos 4t$$

$$\therefore x(t) = e^{-4t} (C_1 + C_2 t) - \frac{1}{4} \cos 4t$$

$$\Rightarrow x'(t) = e^{-4t} (-4C_1 + C_2 - 4C_2 t) + \sin 4t$$

$$x(0) = 0$$

$$\Rightarrow C_1 - \frac{1}{4} = 0$$

$$\Rightarrow C_1 = \frac{1}{4}$$

$$\text{then, } x'(0) = 0$$

$$\Rightarrow 1(-1 + C_2 - 0) + 0 = 0$$

$$\Rightarrow C_2 = 1$$

$$\therefore x(t) = e^{-4t} \left(\frac{1}{4} + t \right) - \frac{1}{4} \cos 4t \quad (\text{Answer})$$

(32) From 31,

$$\text{C.F: } x_c = e^{-4t} (C_1 + C_2 t)$$

$$\text{P.S: } x_p(t) = A e^{-t} \sin 4t + B e^{-t} \cos 4t$$

$$\rightarrow x_p'(t) = -Ae^{-t} \sin 4t + 4Ae^{-t} \cos 4t - Be^{-t} \cos 4t - 4Be^{-t} \sin 4t$$

$$\rightarrow x_p''(t) = Ae^{-t} \sin 4t - 8Ae^{-t} \sin 4t - 16Ae^{-t} \sin 4t + Be^{-t} \cos 4t + 8Be^{-t} \sin 4t - 16Be^{-t} \cos 4t$$

$$\frac{d^2x}{dt^2} + 8 \frac{dx}{dt} + 16x = e^{-t} \sin 4t$$

$$\hookrightarrow u^2 + 8u + 16 = 0$$

$$\hookrightarrow Ae^{-t} \sin 4t - 8Ae^{-t} \sin 4t - 16Ae^{-t} \sin 4t + Be^{-t} \cos 4t + 8Be^{-t} \sin 4t - 16Be^{-t} \cos 4t +$$

$$-8Ae^{-t} \sin 4t + 32Ae^{-t} \cos 4t - 8Be^{-t} \cos 4t - 32Be^{-t} \sin 4t + Ae^{-t} \sin 4t + Be^{-t} \cos 4t = e^{-t} \sin 4t$$

$$\hookrightarrow (A + 8B - 8A - 32B) e^{-t} \sin 4t + (-8A + B + 32A - 8B) e^{-t} \cos 4t = e^{-t} \sin 4t$$

$$\hookrightarrow \left. \begin{array}{l} -7A - 24B = 1 \\ 24A - 7B = 0 \end{array} \right\} \begin{array}{l} A = -\frac{7}{625} \\ B = -\frac{24}{625} \end{array}$$

$$x(t) = C_1 e^{-4t} + C_2 t e^{-4t} - \frac{e^{-t}}{625} (7 \sin 4t + 24 \cos 4t)$$

$$\hookrightarrow x'(t) = -4C_1 e^{-4t} - 4C_2 e^{-4t} + C_2 e^{-4t} + \frac{7}{625} e^{-t} \sin 4t$$

$$- \frac{28}{625} e^{-t} \cos 4t + \frac{24}{625} e^{-t} \cos 4t + \frac{86}{625} e^{-t} \sin 4t$$

$$\hookrightarrow x(0) = C_1 - \frac{24}{625} = 0 \Rightarrow C_1 = \frac{24}{625}$$

$$\hookrightarrow x(0) = C_1 - \frac{24}{625} = 0 \Rightarrow C_1 = \frac{24}{625}$$

$$\hookrightarrow x'(0) = -4C_1 + C_2 - \frac{28}{625} + \frac{24}{625} = 0$$

$$\hookrightarrow C_2 = \frac{4}{25}$$

$$\therefore x(t) = \frac{24}{625} e^{-4t} + \frac{4}{25} t e^{-4t} - \frac{7}{625} e^{-t} \sin 4t - \frac{24}{625} e^{-t} \cos 4t$$

$$\lim_{t \rightarrow \infty} x(t) = 0$$

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$$m = 2 \text{ kg}$$

$$k = 32 \text{ Nm}^{-1}, \quad \beta = 0$$

$$f(t) = 68 e^{-2t} \cos 4t$$

$$\text{Now, } \frac{d^2 x}{dt^2} + \frac{\beta}{m} \frac{dx}{dt} + \frac{k}{m} x = f(t)$$

$$\Rightarrow \frac{d^2 x}{dt^2} + 16x = 68 e^{-2t} \cos 4t$$

$$\therefore \text{CF: } \frac{d^2 x}{dt^2} + 16x = 0$$

$$\hookrightarrow m^2 + 16 = 0$$

$$\Rightarrow \omega = \pm 4i$$

$$\text{So } x_c(t) = C_1 \cos 4t + C_2 \sin 4t$$

$$\text{Now, } x_p(t) = e^{-2t} (A \cos 4t + B \sin 4t)$$

$$\begin{aligned} \Rightarrow x_p'(t) &= -4e^{-2t} A \sin 4t - 2e^{-2t} A \cos 4t + 4Be^{-2t} \cos 4t - 2Be^{-2t} \sin 4t \\ &= e^{-2t} [(-4B - 2A) \cos 4t + (-4A - 2B) \sin 4t] \end{aligned}$$

$$\begin{aligned} \Rightarrow x_p''(t) &= 8e^{-2t} A \sin 4t - 16e^{-2t} A \cos 4t + 4e^{-2t} A \cos 4t + 8e^{-2t} \sin 4t \\ &\quad - 8e^{-2t} \cos 4t - 16Be^{-2t} \sin 4t + 4Be^{-2t} \sin 4t - 8Be^{-2t} \cos 4t \\ &= e^{-2t} [(-12A - 16B) \cos 4t + (16A - 12B) \sin 4t] \end{aligned}$$

$$\text{Now, } \omega^2 + 16 = 0$$

$$\Rightarrow e^{-2t} [(-12A - 16B) \cos 4t + (16A - 12B) \sin 4t] + 16e^{-2t} (A \cos 4t + B \sin 4t) = 68e^{-2t} \cos 4t$$

From coefficient

$$\left. \begin{aligned} 8A - 32B &= 68 \\ 32A + 8B &= 0 \end{aligned} \right\} \begin{aligned} A &= \frac{1}{2} \\ B &= -2 \end{aligned}$$

$$\therefore x_p(t) = e^{-2t} \left(\frac{1}{2} \cos 4t - 2 \sin 4t \right)$$

$$\therefore x(t) = c_1 \cos 4t + c_2 \sin 4t + e^{-2t} \left(\frac{1}{2} \cos 4t - 2 \sin 4t \right)$$

$$\Rightarrow x'(t) = -4c_1 \sin 4t + 4c_2 \cos 4t + e^{2t} (-2 \sin 4t - 8 \cos 4t) - 2e^{2t} \left(\frac{1}{2} \cos 4t - 2 \sin 4t \right)$$

$$\hookrightarrow x(0) = c_1 + 0 + \frac{1}{2} = 0 \Rightarrow C_1 = -\frac{1}{2}$$

$$\hookrightarrow x'(0) = 4c_2 - 8 - c_2 = 0 \Rightarrow C_2 = \frac{9}{4}$$

$$\therefore x(t) = -\frac{1}{2} \cos 4t + \frac{9}{4} \sin 4t + e^{-2t} \left(\frac{1}{2} \cos 4t - 2 \sin 4t \right)$$